SUPPLEMENTARY FILE2:
SUPPLEMENTARY TABLES
S1 TABLE: BRCA1 variant type in studies of South Asia

Author	Year	Number of patients	Age of diagnosis	Country of origin	Clinical phenotype	Breast cancer variant	Method of mutation detection	Exon/ Intron	Nucleotide change (common name)	HGVS Genomic (hg19)	AA change	Variant Type	Frequency	Age of onset (In Years)	Significance
Akter [21]	2019	32	30 to 70 years	Bangladesh	BC	NA	Sanger sequencing	3	c.116G>A	c.116G>A	p.Cys39Tyr	Nonsynonymous	1	NA	Pathogenic mutation
								16	c.5011T>C	c.5011T>C	p.Met1671Val	Nonsynonymous	1	NA	Pathogenic mutation
								10	c.2286A>T	c.2286A>T	p.Arg762Ser	Nonsynonymous	1	NA	VUS
								10	c.3995C>A	c.3995C>A	p.Gly1332Val	Nonsynonymous	1	NA	VUS
								10	c.1058G>A	c.1058G>A	p.Trp353Ter	Stop gain	1	NA	Pathogenic mutation
Silva[38]	2008	130	<40->40 years	Srilanka	BC	Familial and Sporadic	SSCP (Single stranded Conformational Analysis)	11	c.856T>G	c.856T>G	Leu>TrP	Missense unclassified variant	1	NA	Pathogenic/ Likely pathogenic
								11	c.3086delT	c.3086delT	Stop999	Frameshift	1	NA	Pathogenic/ Likely pathogenic
								21	c.5404delG	c.5404delG	Stop1764	Frameshift	1	NA	Pathogenic/ Likely pathogenic
								18	c.5075-2A>T (IVS17-2A>T)	c.5075-2A>T	NA	Intronic variants novel	1	NA	Pathogenic/ Likely pathogenic
								7	c.441+36C>T (IVS7+36C>T)	c.441+36C>T	NA	Unclassified intronic variant	1	NA	VUS
								7	IVS7+38T>C	NA	NA	Unclassified intronic variant	1	NA	Unclassified intronic variant/polymorphism
								7	c.441+41C>T (IVS7+41C>T)	c.441+41C>T	NA	Unclassified intronic variant	1	NA	VUS
								7	IVS7+49del15	NA	NA	Unclassified intronic variant	1	NA	VUS
								8	c.442-34C>T (IVS7-34C>T)	c.442-34C>T	NA	Intronic variants novel	1	NA	VUS

								16	c.4931A>G	c.4931A>G	p.(Glu1644Gly)	Silent-Polymorphism	1	NA	VUS
								16	c.5075G>A	c.5075G>A	Met>Ile	Misense alteration polymorphism	1	NA	Polymorphism/Likely benign
								16	c.4956A>G	c.4956A>G	Ser>Gly	Misense alteration polymorphism	1	NA	Polymorphism/Benign
								11	c.2196G>A	c.2196G>A	Asp > Asn	Misense alteration polymorphism	1	NA	Polymorphism
								11	c.2201C>T	c.2201C>T	Ser>Ser	Silent-Polymorphism	1	NA	Polymorphism
								11	c.2430T>C	c.2430T>C	Leu>Leu	Silent-Polymorphism	1	NA	Polymorphism
								11	c.2731C>T	c.2731C>T	Pro>Leu	Misense alteration polymorphism	1	NA	Polymorphism
								11	c.3232A>G	c.3232A>G	Glu>Gly	Misense alteration polymorphism	1	NA	Polymorphism
								11	c.3667A>G	c.3232A>G	Lys>Arg	Misense alteration polymorphism	1	NA	Polymorphism
								13	c.4427T>C	c.4427T>C	Ser>Ser	Silent-Polymorphism	1	NA	Polymorphism
Liede[35]	2002	341	21 to 81 years	Pakistan	BC	Invasive breast cancer	Direct DNA Sequencing, PTT	NA	c.4485-1G>A (IVS14-1G>A)	c.4485-1G>A	NA	single nucleotide variant	2	35.5	Pathogenic/Likely pathogenic mutation
								11	c.1961dup (2080insA)	c.1961dup	p.Tyr655fs	Duplication	2	35.5	Pathogenic mutation
								11	c.1793T>G (1912T>G)	c.1793T>G	p.(Leu598Ter)	single nucleotide variant	1	45	Pathogenic mutation
								2	c.68_69del (185delAG)	c.68_69del	p.(Glu23ValfsTer17)	NA	1	47.9	Pathogenic mutation
								11	c.1501_1504del (1616del4)	c.1501_1504del	p.(Lys501Ter)	NA	1	35.6	Pathogenic mutation
								12	c.4165_4166del (4284delAG)	c.4165_4166del	p.(Ser1389Ter)	NA	2	39.4	Pathogenic mutation
								11	c.4065_4068del	c.4065_4068del	(p.Asn1355fs)	Deletion	4	35.1	Pathogenic mutation

									(4184del 4)					
								11	1476delG	NA	NA	NA	1	37	NA
								11	c.1813dup (2041insA)	c.1813dup	p.(Ile605AsnfsTer11	Duplication	1	21	Pathogenic mutation
								11	c.3770_3771del (3889delAG)	c.3770_3771del	p.Glu1257fs)	Frameshift mutation	1	38.5	Pathogenic mutation
Moatter[37]	2011	53	28 to 48 years	Pakistan	BC	NA	SSCP assay and DNA sequencing, PTT	11	c.1123T>G	c.1123T > G	stop at 374	Frameshift Mutation	1	28	NA
								20	c.5231delG	c.5231delG	(p.Arg1744fs)	Frameshift Mutation	1	29	Pathogenic mutation
								16	c.4837A>G (4956A>G)	c.4837A > G	p.Ser1613Gly	Missense mutation	1	39	Polymorphism/Benign
								6	c.271T>G	c.271T > G	p.(Cys91Gly)	Missense mutation	1	48	VUS
Rashid[36]	2006	total:126 families bc=39	30-73 years	Pakistan	BC	NA	SSCP, PTT, DHPLC and DNA sequencing	2	c.68_69del (185delAG)	c.68_69del	Stop 39	Frameshift Mutation	2 families	22	Pathogenic mutation
								20	c.5257dup (5376insA)	c.5257dup	p.Arg1753fs	Frameshift Mutation	1 family	22	Pathogenic mutation
								24	c.5503C>T (5622C>T)	c.5503C>T	p.(Arg1835Ter	Nonsense mutation	2 families	25,34,36,37,40,48	Pathogenic mutation
								15	c.4508C>A (4627C>A)	c.4508C>A	p.(Ser1503Ter)	Nonsense mutation	5 families	34,36,37,40,48,34,39,25,31	Pathogenic mutation
								17	c.5030_5033del (5149del4)	c.5030_5033del	p.Thr1677fs	Frameshift Mutation	1 family	28	Pathogenic mutation
								Intron 20	c.5278-1G>C (IVS20-1G>C)	c.5278-1G>C	NA	Splice site mutation	1 family	29	Pathogenic mutation
								12	c.4183C>T (4302C>T)	c.4183C>T	p.(Gln1395Ter)	Nonsense mutation	1 family	30	Pathogenic mutation
								Intron 4	c.135-1G>T (IVS4-1G >T)	c.135-1G>T	NA	Splice site mutation	1 family	32	Pathogenic mutation
								7	c.431del (550delA)	c.431del	(p.Asn144fs)	Frameshift Mutation	1 family	22	Pathogenic mutation

								8	c.470_471del (589delCT)	c.470_471del	p.(Ser157Ter)	Frameshift Mutation	1 family	28,34,35	Pathogenic mutation
Rashid [15]	2019	Total BC families :447	18-78 years	Pakistan	BC	NA	DHPLC	2	c.66_67insA (185insA)	c.66_67insA	p.(Glu23Argfs*18)	Frameshift Mutation	1 family	30,53	Pathogenic mutation
								2	c.68_69del (185delAG)	c.68_69del	p.(Glu23Valfs*17)	Frameshift Mutation	1 family	28,40	Pathogenic mutation
								4	c.135-2A>G (IVS4-2A>G)	c.135-2A>G	NA	Splice site mutation	1 family	30	Pathogenic mutation
								7	c.335del (454delA)	c.335del	p.(Asn112Ilefs*7)	Frameshift Mutation	1 family	27,32,43	Pathogenic mutation
								7	c.390C>G (Y130X)	c.390C>G	p.Tyr130Ter)	Nonsense mutation	1 family	32,55	Pathogenic mutation
								11	c.685del (804delT)	c.685del	p.(Ser229Leufs*5)	Frameshift Mutation	7 families	21,<30,34,43,44,51,52,41,46,34,42,<50,<50,<50,<50,30, <32, 25, 27, 33, 70, 26, 35, 21, <30, 34, 43, 44, 51,52	Pathogenic mutation
								11	c.784C>T (Q262X)	c.784C>T	p.Gln262Ter	Nonsense mutation	1 family	40,40	Pathogenic mutation
								11	c.895_896del (1014delGT)	c.895_896del	p.Val299fs)	Frameshift Mutation	1 family	40,NA,NA	Pathogenic mutation
								11	c.1008del (1127delA)	c.1008del	p.(Glu337Lysfs*4)	Frameshift Mutation	1 family	25,<40,<50	Pathogenic mutation
								11	c.1188del (1307delT)	c.1188del	p.(Asp396Glufs*14)	Frameshift Mutation	1 family	47	Pathogenic mutation
								11	c.1190del (1309delA)	c.1190del	p.(Asp397Alafs*13)	Frameshift Mutation	1 family	29,64	Pathogenic mutation
								11	c.1399_1453dup (1518_1572dup55)	c.1399_1453dup	p.(Ala485Glufs*13)	Frameshift Mutation	1 family	20,34	Pathogenic mutation
								11	c.1471C>T (1590C>T)	c.1471C>T	p.Gln491Ter	Nonsense mutation	1 family	23,30,38	Pathogenic mutation

	11	c.1779_1785del (1898delTATGGAA)	c.1779_1785de	p.(Met594Serfs*3)	Frameshift Mutation	1 family	26	Pathogenic mutation
	11	c.1793T>G (1912T>G)	c.1793T>G	p.(Leu598*)	Nonsense mutation	3 families	30,29,30,31,39,32,45	Pathogenic mutation
	11	c.1961dup (2080insA)	c.1961dup	p.(Tyr655Valfs*18)	Frameshift Mutation	1 family	32,32,35,45,48,<50	Pathogenic mutation
	11	c.2149G>T (E717X)	c.2149G>T	p.(Glu717*)	Nonsense mutation	1 family	40,55	Pathogenic mutation
	11	c.2269del (2388delG)	c.2269del	p.(Val757Phefs*8)	Frameshift Mutation	5 families	22,34,56,25,42,45,35,55,<55,43,44	Pathogenic mutation
	11	c.2314_2315del (2433delGT)	c.2314_2315del	p.(Val772Thrfs*4)	Frameshift Mutation	1 family	32,35,70	Pathogenic mutation
	11	c.2340_2343del (2459delGGAA)	c.2340_2343del	p.(Glu781Valfs*10)	Frameshift Mutation	2 families	31,32,35,45,29,29,<35,>35,>55	Pathogenic mutation
	11	c.2405_2406del (2524delTG)	c.2405_2406del	p.(Val802Glufs*7)	Frameshift Mutation	5 families	30,33,34,50,40,55,40,30	Pathogenic mutation
	11	c.2538_2540delinsG (2657delAAT-insG)	c.2538_2540delinsG	p.(Met847Glyfs*4)	Frameshift Mutation	1 family	24,<30,31	Pathogenic mutation
	11	c.2603C>G (S868X)	c.2603C>G	p.(Ser868*)	Nonsense mutation	2 families	25,70,34,35	Pathogenic mutation
	11	c.2971A>T (K991X)	c.2971A>T	p.(Lys991*)	Nonsense mutation	1 family	29	Pathogenic mutation
	11	c.2981_2982del (3100delGT)	c.2981_2982del	p.(Cys994*)	Frameshift Mutation	1 family	27,28,40,42	Pathogenic mutation
	11	c.3129_3138del (3248delTATTAATGAA)	c.3129_3138del	p.(Asn1043Lysfs*2)	Frameshift Mutation	1 family	24,55	Pathogenic mutation
	11	c.3339_3341del (3458delTGA)	c.3339_3341del	p.(Tyr1113*)	Frameshift Mutation	2 families	43,44,50,NA,40,>50	Pathogenic mutation
	11	c.3412G>T (G1138X)	c.3412G>T	p.(Gly1138*)	Nonsense mutation	1 family	28	Pathogenic mutation

11	c.3477_3480del (3596del4)	c.3477_3480del	p.(Ile1159Metfs*50)	Frameshift Mutation	1 family	28	Pathogenic mutation
11	c.3598C>T (3717C>T)	c.3598C>T	p.(Gln1200*)	Nonsense mutation	2 families	27,44,>60,73	Pathogenic mutation
11	c.3607C>T (3726C>T)	c.3607C>T	p.(Arg1203*)	Nonsense mutation	1 family	36,50,NA	Pathogenic mutation
11	c.3700_3704del (3819del5)	c.3700_3704del	p.(Val1234Glnfs*8)	Frameshift Mutation	1 family	34	Pathogenic mutation
11	c.3770_3771del (3889delAG)	c.3770_3771del	p.(Glu1257Glyfs*9)	Frameshift Mutation	10 families	45,26,50,30,48,NA,25,26,51,35,58,30,47,27,31,33,42,50,24,32,35,37,50	Pathogenic mutation
11	c.4065_4068del (4184del4)	c.4065_4068del	(p.Asn1355fs)	Frameshift Mutation	4 families	29,41,45,40,44,62,28,40,NA	Pathogenic mutation
11	c.4183C>T(4302C>T)	c.4183C>T	p.(Gln1395*)	Nonsense mutation	1 family	40,46	Pathogenic mutation
11	c.4327C>T (4446C>T)	c.4327C>T	p.(Arg1443*)	Nonsense mutation	1 family	24,31,33,50	Pathogenic mutation
14	c.4485-1G>A(IVS14-1G>A)	c.4485-1G>A	NA	Splice site mutation	6 families	29,36,39,40,53,29,<50,29,<30,40,30,56,50,>60,29,<30,40	Pathogenic/ Likely pathogenic mutation
15	c.4508C>A (4627C>A)	c.4508C>A	p.(Ser1503*)	Nonsense mutation	3 families	25,27,<30,53,58,63,77,<25,<30,<40,46,<50,>50,34,40,42,50	Pathogenic mutation
15	c.4665del (4784delG)	c.4665del	p.(Arg1555Serfs*4)	Frameshift Mutation	1 family	29,31,48,65	Pathogenic mutation
15	c.4862del (4981delA)	c.4862del	p.(Asp1621Valfs*12)	Frameshift Mutation	1 family	31,36,37,42	Pathogenic mutation
15	c.5035del (5154delC)	c.5035del	p.(Leu1679*)	Frameshift Mutation	2 families	30,37,45,NA,NA,NA,NA,NA,26,44,50	Pathogenic mutation
17	c.5074+1G>A (IVS17+1G>A)	c.5074+1G>A	NA	Splice site mutation	2 families	28,28,<33	Pathogenic mutation

								20	c.5239del (5358delC)	c.5239del	p.(Gln1747Lysfs*18)	Frameshift Mutation	1 family	26,26,NA	Pathogenic mutation
								20	c.5266dup (5385dupC)	c.5266dup	p.(Gln1756Profs*74)	Frameshift Mutation	1 family	37,38,55,NA,NA	Pathogenic mutation
								20	c.5278-1G>C (IVS20-1G>C)	c.5278-1G>C	NA	Splice site mutation	1 family	26	Pathogenic mutation
								21	c.5310dup (5429dupG)	c.5310dup	p.(Pro1771Alafs*59)	Frameshift Mutation	1 family	28	Pathogenic mutation
								22	c.5361_5362del (5480delTG)	c.5361_5362del	p.(Cys1787Trpfs*42)	Frameshift Mutation	2 families	25,32,30,40,44,48	Pathogenic mutation
								22	c.5377A>T (4446C>T)	c.5377A>T	p.(Lys1793*)	Nonsense mutation	1 family	28,40	Pathogenic mutation
								23	c.5468-2A>T (IVS23-2A>T)	c.5468-2A>T	NA	Splice site mutation	1 family	23,<25,34,<40,<46,<60,NA	Pathogenic/ Likely pathogenic mutation
								24	c.5503C>T (5622C>T)	c.5503C>T	p.(Arg1835*)	Nonsense mutation	9 families	25,28,26,34,39,NA,31,50,85,22,42,48,<25,47,34,35,>50,NA,<25,71,30,<40,45,45	Pathogenic mutation
								1-2	c.(?_-1)_(80+1_81-1)del(del exon 1-2)	c.(?_-1)_(80+1_81-1)del	NA	Large genomic rearrangement	7 families	28,29,31,39,40,40,41,38,72,32,42,56,70,NA,NA39,42,48	NA
								21-24	del exon 21-24	NA	NA	Large genomic rearrangement	2 families	33,34,>40,42,NA	NA
Darooei M[32]	2017	22	29-83 years	India	BC	HBOC	Next generation sequencing	16	c.5074+1G>A (IVS17+1G>A)	c.5074+1G>A	NA	NA	1	35	Pathogenic mutation
								10	c.2426A>G	c.2426A>G	p.Glu809Gly	NA	1	39	VUS
								2	c.68_69del (185delAG)	c.68_69del	p.Glu23Val	NA	1	41	Pathogenic mutation
								12	c.4224_4231del	c.4224_4231del	p.Gln1408His	Frameshift mutation	1	48	Pathogenic mutation
								2	c.68del (187delA)	c.68del	p.Glu23Val	Frameshift mutation	1	83	Pathogenic mutation

								16	c.5074+1G>A (IVS17+1G>A)	c.5074+1G>A	NA	NA	1	49	Pathogenic mutation
								10	c.1504_1508del (1623del5)	c.1504_1508del	p.Leu502Ala	NA	1	28	Pathogenic mutation
								14	c.4485-1G>A (IVS14-1G>A)	NA	NA	NA	1	55	Pathogenic mutation
Hedau [24]	2004	124	27-79 years	India	BC	sporadic and familial	`	2	T131G	NA	NA	Transition silent	1	60	NA
								2	A128G	NA	NA	Transversion silent	1	38	NA
								16	4956insG	NA	TGA at codon 1621	Frameshift mutation	1	45	NA
								12	c.4183C>T (4302C>T)	c.4183C>T	Gln1395Stop	Transition nonsense	1	40	Pathogenic mutation
								7	A448C	NA	Lys110Thr	Transversion missense	1	40	NA
								7	T459C	NA	Ser114Pro	Transition missense	1	35	NA
								7	c.442-34C>T (IVS7-34C>T)	c.442-34C>T	NA	Transition substitution	5	37	Benign or VUS according to BIC
								2	c.68_69del (185delAG)	c.68_69del	TGA at codon 39	Frameshift mutation	2	37	Pathogenic mutation
								18	c.5152+66G>A (5271+66G>A)	c.5152+66G>A	NA	Transition substitution	3	40	Benign
								13	T4427C	NA	NA	Transition silent	4	48	NA
Juwle[29]	2012	50	28-45 years	India	BC	Early onset BC	PCR	2	c.68_69del (185delAG)	c.68_69del	p.(Glu23ValfsTer17)	Frameshift mutation	2	NA	Pathogenic mutation
								8	632insT		Stop181	Frameshift mutation	2	NA	Pathogenic mutation
								11	c.2275C>T (Q759X)	c.2275C>T	p.(Gln759Ter)	Nonsense mutation	1	NA	Pathogenic mutation
								11	c.2338C>T (2457C>T)	c.2338C>T	p.(Gln780Ter)	Nonsense mutation	2	NA	Pathogenic mutation
								11	c.3607C>T	c.3607C>T	p.(Arg1203Ter)	Nonsense mutation	1	NA	Pathogenic mutation

									(3726C>T)						
								14	c.4485-1G>A (IVS14-1G>A)	c.4485-1G>A	NA	Splice Site (IVS)	1	NA	Pathogenic/ Likely pathogenic
								17	c.5074+1G>T (IVS17+1G>T)	c.5074+1G>T	NA	Splice Site (IVS)	1	NA	Pathogenic/ Likely pathogenic
								11	c.823G>A (942G>A)	c.823G>A	G275S	Missense mutation	1	NA	NA
								11	c.3119G>A (3238G>A)	c.3119G>A	p.(Ser1040Asn)	Missense mutation	1	NA	Benign
								16	c.4883T>C (5002T>C)	c.4883T>C	p.(Met1628Thr)	Missense mutation	1	NA	Benign
								16	c.4957G>A (5076G>A)	c.4957G>A	p.(Val1653Met)	Missense mutation	6	NA	VUS
Kumar [22]	2002	14	NA	India	BC	NA	Direct Sequencing of PCR	7	465G>A	NA	Glu to Lys (G-A)	Missense mutation	1	NA	NA
								11	1027delA	NA	Stop codon 313	Frameshift mutation	1	NA	NA
								2	c.68_69del (185delAG)	c.68_69del	Stop codon 39	Frameshift mutation	1	NA	Pathogenic mutation
Maanan[31]	2016	91	26 to 75 years	India	BC	familial and sporadic	Multi-gene sequencing	1,2	c.(?_-119)_(80+1_81-1)del	c.(?_-119)_(80+1_81-1)del	NA	large deletion	1	35	NA
								2	c.68_69del (185delAG)	c.68_69delAG	p.Glu23ValfsTer17	Small insertion deletion	6	62,39,48, 41,32,33	Pathogenic mutation
								9	c.671-1G>T (IVS10-1G>T)	c.671-1G>T	NA	Splice Site	1	38	Pathogenic mutation
								10	c.779dupA	c.779dupA	p.Tyr261ValfsTer4	Small insertion deletion	1	40	NA
								10	c.3352C>T (3471C>T)	c.3352C>T	p.Gln1118Ter	Nonsense mutation	2	48,34	Pathogenic mutation
								10	c.4065_4068del (4184del4)	c.4065_4068del	p.Asn1355LysfsTer10	Small insertion deletion	1	42	Pathogenic mutation
								10	c.809del (928delA)	c.809del	p.His270LeufsTer28	Small insertion deletion	1	54	Pathogenic mutation

								10	c.1155G>A	c.1155G>A	p.Trp385Ter	Nonsense mutation	1	40	Pathogenic mutation
								10	c.1416delC	c.1416del	p.Asn473ThrfsTer2	Small insertion deletion/frameshift	1	33	Pathogenic mutation
								11	c.4183C>T (4302C>T)	c.4183C>T	p.Gln1395Ter	Nonsense mutation	1	46	Pathogenic mutation
								11	c.4165_4166del (4284delAG)	c.4165_4166del	p.Ser1389TerfsTer1	Small insertion deletion	1	29	Pathogenic mutation
								13	c.4484+1G>A	c.4484+1G>A	NA	Splice site	1	49	Pathogenic mutation
								15	c.4837_4838delinsGCC (S1566fs)	c.4837_4838delinsGCC	p.Ser1613AlafsTer9	Small insertion deletion	2	37	Pathogenic mutation
								16	c.5035del (5154delC)	c.5035del	p.Leu1679TerfsTer1	Small insertion deletion	1	36	Pathogenic mutation
								16	c.5074+1G>A (IVS17+1G>A)	c.5074+1G>A	NA	Splice site	3	45,49	Pathogenic mutation
								17	c.5133del (K1711fs)	c.5133del	p.Lys1711AsnfsTer2	Small insertion deletion	1	50	Pathogenic mutation
								22	c.5440dup (A1814fs)	c.5440dup	p.Ala1814GlyfsTer16	Small insertion deletion	1	27	Pathogenic/ Likely pathogenic
								22	c.5467+1G>A (IVS23+1G>A)	c.5467+1G>A	NA	Splice site	1	24	Pathogenic mutation
Mehta [33]	2018	126	<50->50 years	India	BC	NA	Next generation sequencing	2	c.68_69del (185delAG)	c.68_69delAG	p.Glu23fs	Frameshift mutation	1	NA	Pathogenic mutation
								7	c.541G>T (E134*)	c.541G>T	p.Lys739Ter	Nonsense mutation	1	NA	Pathogenic mutation
								10	c.2214_2215insT	NA	p.Lys739Ter	Nonsense mutation	2	NA	NA
								10	c.2269del (2388delG)	c.2269del	p.Val757fs	Frameshift mutation	1	NA	Pathogenic mutation
								10	c.2295delG (S766fs)	c.2295del	p.Ser766fs	Frameshift mutation	1	NA	Likely pathogenic
								10	c.2362delG (V788fs)	c.2362del	p.Val788fs	Frameshift mutation	1	NA	Pathogenic mutation
								10	c.3328-3330delAAG (3447del3)	c.3328-3330del	p.Lys1110del	Missense mutation	1	NA	VUS

								11	c.4108delT (S1370fs)	c.4108del	p.Ser1370fs	Frameshift mutation	1	NA	NA
								11	c.4158_4162delCTCTC (4277del5)	c.4158_4162del	p.Ser1387fs	Frameshift mutation	2	NA	Pathogenic mutation
								12	c.4328G>A (R1443Q)	c.4328G>A	p.Arg1443Gln	Missense mutation	1	NA	VUS
								15	c.4508C>A (4627C>A)	c.4508C>A	p.Ser1524Ter	Nonsense mutation	2	NA	Pathogenic mutation
								17	Splicesite_3 (C>T)	NA	NA	Splicing	1	NA	NA
								21	c.5278-1G>C (IVS20-1G>C)	c.5278-1G>C	NA	Splicing	1	NA	Pathogenic mutation
								21-23	Duplication (exons 21–23)	NA	NA	Big duplication	2	NA	NA
								23	Deletion (exon 23)	NA	Truncated protein codon 1770	Big deletion	1	NA	NA
								23	c.5465G>A (G1822D)	c.5402G>A	p.Gly1822Asp	Missense mutation	1	NA	Benign
								24	Deletion (exon 24)	NA	NA	Big deletion	1	NA	NA
Saxena[25]	2006	196	13 to 78 years	India	Breast cancer	Early onset, late onset , familial	heteroduplex analysis , direct sequencing	2	c.68_69del(185delAG)	c.68_69del	Stop 39	Frameshift mutation	1	40	Pathogenic mutation
								11	c.4065_4068del (4184del4)	c.4065_4068del	(p.Asn1355fs)	Frameshift mutation	1	60	Pathogenic mutation
								11	c.3477_3480del (3596del4)	c.3477_3480del	p.(Ile1159MetfsTer50)	Frameshift mutation	1	24	Pathogenic mutation
								1	22C>G	NA	NA	Transition UV	1	35	Novel mutation
								7	IVS7+38T>C	NA	NA	Transversion Intronic, UV	1	30	Novel mutation
								11	c.671-12del (IVS10-12delG)	c.671-12del	NA	Deletion Intronic, UV	4	35,35,35,31	Novel mutation
								11	c.3548_3549delinsGG(K1183R)	c.3548_3549delinsGG	p.Lys1183Arg	Polymorphism	16	NA	VUS

								13	c.4357+2T>C (IVS13+2 T>C)	c.4357+2T>C	NA	Transversion Intronic, UV	1	30	Pathogenic mutation
								16	c.4837A>G(4956A>G)	c.4837A>G	p.(Ser1613Gly)	Polymorphism	1	62	Benign
								16	c.4956G>A(5075G>A)	c.4956G>A	p.(Met1652Ile)	Polymorphism	14	NA	Benign
								17	c.5000A>G(K1667R)	c.5000A>G	p.(Lys1667Arg)	Transversion, Missense UV	1	35	VUS
								17	5154 C>T	NA	Leu to Leu	Transversion Silent, UV	1	NA	NA
Shah[34]	2018	79	23-74 years	India	BC	NA	NGS analysis	2	c.66_67delAG	c.66_67delAG	p.Glu23Val	Frameshift mutation	1	51	Pathogenic mutation
								10	c.3328_3330del (K1110del)	c.3328_3330del	p.Lys1110del	In frame deletion	1	30	VUS
								14	c.4442_4442delG	c.4442_4442del	p.Ser1481fs	Frameshift mutation	1	37	Novel mutation
Vaidyananth [27]	2009	39	NA	India	BC	Familial BC	Direct sequencing	2	c.68_69del(185delAG)	c.68_69del	Stop codon 39	Frameshift mutation	6	NA	Pathogenic mutation
								11	c.2864C>A (2983C>A)	c.2864C>A	p.(Ser955Ter)	Nonsense mutation	1	NA	Pathogenic mutation
								11	c.4094del (4213delT)	c.4094del	p.(Leu1365Ter)	Nonsense mutation	1	NA	Pathogenic mutation
Singh[30]	2015	381	<40->40	India	BC	NA	PCR-SSCP	2	c.66_67delAG	c.66_67delAG	Stop 39	Frameshift mutation	2	28,46	Pathogenic mutation
								11	1230delC	NA	Stop 373	Frameshift mutation	1	30	Novel mutation
								11	1238delA	NA	Stop 375	Frameshift mutation	1	30	Novel mutation
								11	1273 insG	NA	Stop 390	Frameshift mutation	1	30	Novel mutation
								11	1415 insA	NA	Stop 435	Frameshift mutation	1	30	Novel mutation
								11	1611 delC	NA	Stop 502	Frameshift mutation	1	30	Novel mutation
								11	1614 insC	NA	Stop 500	Frameshift mutation	1	30	Novel mutation

								11	2025 ins C	NA	Stop 638	Frameshift mutation	1	40	Novel mutation
								11	2361insC	NA	Stop 761	Frameshift mutation	1	40	Novel mutation
								11	1961 del G	NA	Stop 625	Frameshift mutation	1	45	Novel mutation
								11	2354 A to C	NA	Arg-Ser	Misense mutation	1	45	Novel mutation
								11	2026 T to A	NA	Leu-stop	Nonsense mutation	2	40,40	Novel mutation
Saxena[23]	2002	19	25-64 Years	India	BC	Familial BC	Heteroduplex analysis, PCR	5	c.212+1 G>T (331+1G >T)	c.212+1G >T	NA	Abrrent splicing	1	30	Pathogenic mutation
								13	c.4357+2 T>C (IVS13+ 2 T>C)	c.4357+2T >C	NA	Abrrent splicing	1	30	Pathogenic mutation
Soumittra[28]	2009	Total: 91 Bc: 71	NA	India	BC	HBOC /HBC/ Early BC	PCR-DHPLC	12	c.4158_4 162delC TCTC (4277del 5)	c.4158_41 62del	p.Ser1369 SerfsX2	NA	1	28	Pathogenic mutation
								13	c.4327C> T (4446C> T)	c.4327C> T	p.(Arg144 3Ter)	NA	1	40	Pathogenic mutation
								11	c.1148_1 149delA T (1267del AT)	c.1148_11 49del	p.Asn383 Arg fsX6	NA	1	38	Pathogenic mutation
								14	c.4399C> T (4518C> T)	c.4399C> T	p.Gln1467 X	NA	1	38	Pathogenic mutation
								16	c.4705_4 706insT GGAAT C	NA	p.Ile1567f sx5	NA	1	45	NA
								2	c.68_69d el (185delA G)	c.68_69del AG	p.Glu23Va l fsX16	Frameshift mutation	3	33,33,39	Pathogenic mutation
								2	c.66_67d elAG	c.66_67del AG	p. Leu22Leu fsX18	Frameshift mutation	1	55	Pathogenic mutation
								18	c.5118_5 120delA AT	c.5118_51 20delAAT	p.del1707I le	NA	1	45	NA

S2 TABLE: BRCA2 variant type in studies of South Asia

Author	Year	Number of patients	Age of diagnosis	Country of origin	Clinical phenotype	Breast cancer variant	Method of mutation detection	Exon/ Intron	Nucleotide change		AA change	Mutation type	Frequency	Age of Onset(years)	Significance
Akter[21]	2019	32	30 to 70 years	Bangladesh	BC	NA	Sanger sequencing	11	c.2459A>G (D820G)	c.2459A>G	p.Asp820Gly	Non synonomous	1	NA	VUS
								16	c.7722G>A	c.7722G>A	p.Trp2574Ter	Stop gain	1	NA	Pathogenic mutation
								10	c.1114A>C (1342 A>C)	c.1114A>C	p.Asn372His	Non synonomous	1	NA	Benign
								10	c.1792A>G (T598A)	c.1792A>G	p.Thr598Ala	Non synonomous	1	NA	Benign
								10	c.1301_1308del (K436fs)	c.1301_1308del	p.Lys436Phefs*	Frameshift deletion	1	NA	Pathogenic mutation
								4	c.351_352delTC	c.351_352del	p.Arg118Hisfs*	Frameshift deletion	1	NA	Pathogenic mutation
								11	c.6451G>A	c.6451G>A	p.Val2151Ile	Non synonomous	1	NA	VUS
de Silva[39]	2017	48	18 to 45 years	Srilanka	BC	NA	Direct sequencing	Exon 11	c.2411_2412del (E804fs)	c.2411_2412del	p.Glu804Valfs*2	Frameshift deletion	1	NA	Pathogenic mutation
								Exon 11	c.2500_2501insG (L834fs)	c.2500_2501insG	p.Leu834Cysfs*4	Frameshift insertion	1	NA	Pathogenic mutation
								Exon 11	c.3881T>G (4109T>G)	c.3881T>G	p.Leu1294*	Nonsense	1	NA	Pathogenic mutation
								Exon 11	c.4768A>T (K1590*)	c.4768A>T	p.Lys1590*	Nonsense	2	NA	Pathogenic mutation

								Exon 11	c.5645C>G (S1882X)	c.5645C>G	p.Ser1882*	Nonsense	2	NA	Pathogenic mutation
								Exon 11	c.5747delC	c.5747delC	p.His1916Leufs*47	Frameshift deletion	1	NA	Pathogenic mutation
								Exon 11	c.6728C>T	c.6728C>T	p.Ser2243Phe	Missense	3	NA	VUS
								Exon 11	c.1922C>T (S641F)	c.1922C>T	p.Ser641Phe	Missense	2	NA	VUS
								Exon 11	c.3378A>T	c.3378A>T	p.Glu1126Asp	Missense	1	NA	Likely pathogenic
								11	c.2324C>T (S775F)	c.2324C>T	p.Ser775Phe	Missense	1	NA	VUS
								11	c.5104C>T (P1702S)	c.5104C>T	p.Pro1702Ser	Missense	1	NA	VUS
Liede A[35]	2002	341	20-81	Pakistan	BC	Invasive BC	PTT	(exon 11)	c.5074_5075insA (5302insA)	c.5074_5075insA	p.(Trp1692Ter)	NA	1	48.4	Pathogenic mutation
								exon 22)	c.8912del (9140delA)	c.8912delA	p.(Lys2971SerfsTer5)	Frameshift	1	40.3	Pathogenic mutation
								Exon 11	c.3109C>T (3337C>T)	c.3109C>T	p.Gln1037Ter	Nonsense	2	39.2,40	Pathogenic mutation
								Exon 11	c.3685delG (3913delG)	c.3685delG	p.(Val1229PhefsTer10)	Frameshift	1	62.1	Pathogenic mutation
								Exon 11	c.5722_5723del (5950delCT)	c.5722_5723del	p.(Leu1908ArgfsTer2)	Frameshift	1	39.7	Pathogenic mutation
								Exon 11	c.6468_6469del (6696delTC)	c.6468_6469del	p.(Gln2157IlefsTer18)	Frameshift	1	61.4	Pathogenic mutation
								Exon 11	c.2446del (2674delG)	c.2446del	p.(Glu816LysfsTer9)	Frameshift	1	39.6	Pathogenic mutation
Rashid[36]	2006	total :126 families bc= 39		Pakistan	BC	NA	SSCP, DHPLC and DNA sequencing	10	c.1855C>T (Q619X)	c.1855C>T	Gln to stop	Nonsense mutation	1 family	30	Pathogenic mutation
								10	c.1765_1766del (1993delAA)	c.1765_1766del	Stop 595	Frameshift mutation	1 family	30	Pathogenic mutation
								11	c.4829_4830del (5057delTG)	c.4829_4830del	Stop 1613	Frameshift mutation	1 family	27,57	Pathogenic mutation
								25	3144delT	NA	Stop 3162	Frameshift mutation	1 family	45,69	NA
								11	c.2990T>G (L992X)	c.2990T>G	Leu to stop	Nonsense mutation	1 family	26,33,38,<50	Pathogenic mutation

								11	c.3824_3827del (4052delTAGA)	c.3824_3827del	Stop 1282	Frameshift mutation	1 family	35	Pathogenic mutation
Rashid[15]	2019	Total:447 families bc=357		Pakistan	BC	NA	DHPLC	3	c.92G>A (W31X)	c.92G>A	p.(Trp31*)	Nonsense mutation	2 families	35,46,51,55,50	Pathogenic mutation
								4	c.426-2A>G (IVS4-2A>G)	c.426-2A>G	NA	Splice site	1 family	26,35	Pathogenic mutation
								9	c.765_769del (993delCACAA)	c.765_769del	p.(Asn255Lysfs*19)	Frameshift mutation	1 family	35,36,47	Pathogenic/Likely pathogenic
								10	c.1300_1303del (1528delAAAA)	c.1300_1303del	p.(Lys434Glufs*25)	Frameshift mutation	1 family	29	Pathogenic mutation
								11	c.2820del (3048delA)	c.2820del	p.(Val941Cysfs*19)	Frameshift mutation	1 family	31,43	Pathogenic mutation
								11	c.2835del (3063delA)	c.2835del	p.(Asp946Ilefs*14)	Frameshift mutation	1 family	30	Pathogenic mutation
								11	c.3860del (4088delA)	c.3860del	p.(Asn1287Ilefs*6)	Frameshift mutation	1 family	43,46	Pathogenic mutation
								11	c.5222_5225del (5450delGTAA)	c.5222_5225del	p.(Ser1741Thrfs*35)	Frameshift mutation	4 families	35,42,43,50,54,60,NA32,33,<50,60,6731,45,54,45	Pathogenic mutation
								11	c.5682C>A (Y1894X)	c.5682C>A	p.(Tyr1894*)	Nonsense mutation	2 families	42,51,40,45,48,50,<57	Pathogenic mutation
								11	c.5722_5723del (5950delCT)	c.5722_5723del	p.(Leu1908Argfs*2)	Frameshift mutation	1 family	23	Pathogenic mutation
								11	c.6468_6469del (6696delTC)	c.6468_6469del	p.(Gln2157Ilefs*18)	Frameshift mutation	1 family	31,<50,55	Pathogenic mutation
								15	c.7575del (7803delA)	c.7575del	p.(Ala2526Glnfs*2)	Frameshift mutation	1 family	45,50,69	Pathogenic mutation
								17	c.7976+2C>A (IVS17+2C>A)	c.7976+2C>A	NA	Splice site	1 family	38,39	Pathogenic mutation
								20	c.8545_8546del (8773delAA)	c.8545_8546del	p.(Lys2849Glyfs*19)	Frameshift mutation	1 family	35,56	Pathogenic mutation

								20	c.8551_8570dup (8779_8798dup20)	c.8551_8570dup	p.(Lys2860Asnfs*10)	Frameshift mutation	1 family	26,30,<33,70	Pathogenic mutation
								21	c.8754+4A>G (IVS21+4A>G)	c.8754+4A>G	NA	Splice site	1 family	34,38,45,50,58	Pathogenic mutation
Darooei M[32]	2017	22	29-83	India	BC	NA	Next Generation Sequencing	Exon 15	c.7445C>T	NA	(p.Thr2482Ile)	NA	1	56	NA
								Exon 11	c.5986G>A (A1996T)	c.5986G>A	(p.Ala1996Thr)	Missense	1	52	VUS
								Exon 20	c.8587dup (E2863fs)	c.8587dup	(p.Glu2863Gly)	Frameshift	1	59	Pathogenic mutation
								Exon 10	c.1792A>G (T598A)	c.1792A>G	p.Thr598Ala	NA	1	47	Benign
Hedau S[24]	2004	124	27-79	India	BC	Sporadic BC and Familial BC	PCR/Direct sequencing	2	G292C	NA	Ala22Pro	Transversion missense	1	40	NA
								2	C282G	NA	NA	Transversion silent	1	30	NA
								2	G203A	NA	NA	Transition substitution	4	52,45,48,29	NA
Juwle[29]	2012	50	28-45 Years	India	BC	NA	PCR	11	c.3847_3848del (4075delGT)	c.3847_3848del	Stop1284	Frameshift	1	NA	Pathogenic
								11	c.5076del (5076delAA)	c.5076del	Stop1617	Frameshift	1	NA	Pathogenic
								11	c.6078_6079del (6079delAGTT)	c.6078_6079del	Stop1961	Frameshift	1	NA	Pathogenic
								25	c.9380G>A (9608G>A)	c.9380G>A	p.(Trp3127Ter)	Nonsense	1	NA	Pathogenic
								10	c.1166C>A (1394C>A)	c.1166C>A	P389Q	Missense	1	NA	Benign
								11	c.3515C>T (3743C>T)	c.3515C>T	S1172L	Missense	1	NA	Benign
								11	c.4779A>C (5007A>C)	c.4779A>C	E1593D	Missense	1	NA	Benign
								11	c.5095G>A (5323G>A)	c.5095G>A	D1699 N	Missense	1	NA	VUS
								11	c.5986G>A (A1996T)	c.5986G>A	A1996T	Missense	1	NA	VUS
								11	c.6323G>A (6551G>A)	c.6323G>A	R2108H	Missense	1	NA	Benign
								13	c.6943A>C (7171A>C)	c.6943A>C	I2315L	Missense	1	NA	VUS
								24	c.9235G>A (9463G>A)	c.9235G>A	V3079I*	Missense	1	NA	Benign
								27	c.10154G>A (10382G>A)	c.10154G>A	R3385H*	Missense	1	NA	VUS

Maanan [31]	2016	91	26 to 75 years	India	BC	Familial or sporadic	Multi-gene sequencing	Exon 3	c.92G>A (W31X)	c.92G>A	p.Trp31Ter	Nonsense mutation	1	35	Pathogenic mutation
								Exon 11	c.3187C>T (3415C>T)	c.3187C>T	p.Gln1063Ter	Nonsense mutation	1	50	Pathogenic mutation
								Exon 11	c.3189_3192del (3417del4)	c.3189_3192del	p.Ser1064LeufsTer12	Small insertion and deletion	1	42	Pathogenic mutation
Mehta A[33]	2018	132	50->50 years	India	BC	NA	NGS	10	c.932_933insT	c.932_933insT	p.Ser313fs	Frameshift	1	NA	Pathogenic mutation
								10	c.950_951insA	c.950_951ins	p.Asn319fs	Frameshift	1	NA	Pathogenic mutation
								11	c.2808_2811del (3034del4)	c.2808_2811del	p.Ala938fs	Frameshift	1	NA	Pathogenic mutation
								11	c.5095G>A (5323G>A)	c.5095G>A	p.Asp1699Asn	Missense	1	NA	VUS
								11	c.6125A>C (Q2042P)	c.6125A>C	p.Gln2042Pro	Missense	1	NA	VUS
								12	c.6886A>C (I2296L)	c.6886A>C	p.Ile2296leu	Missense	1	NA	VUS
								14	c.7301A>C (K2434T)	c.7301A>C	p.Lys2434Thr	Missense	1	NA	VUS
								27	c.9976A>T (10204A>T)	c.9976A>T	p.Lys3326Ter	Nonsense	1	NA	Benign
Saxena S[25]	2006	204	13-78 years	India	BC	Early onset, late onset, familial	PCR	11	6376InsAA	NA	Stop2051	Frameshift	1	30	Pathogenic mutation
								19	8576InsC	NA	Stop2797	Frameshift	1	35	Pathogenic mutation
								27B	9999delA	NA	Stop3275	Frameshift	1	50	Pathogenic mutation
Shah N[34]	2018	79	23-72	India	BC	NA	NGS/Sanger Sequencing	Exon 6	c.484_484delG	c.484_484delG	p.Pro163fs	Frameshift deletion	1	29	Novel mutation
								Exon 11	c.3167_3170del (3395del4)	c.3167_3170del	p.Gln1056	Frameshift deletion	1	48	Pathogenic mutation
								Exon 10	c.2169_2170insA	c.2169_2170ins	p.Val726fs	Frameshift Insertion	1	50	Novel mutation
Syamala [26]	2007	94	22-79	India	BC	NA	Direct sequencing of PCR products	11	c.4642delAA	c.4642delAA	STOP 1480	Frameshift mutation	1	43	Pathogenic mutation

								11	c.4926insGACC	c.4926insGACC	STOP 1575	Frameshift mutation	1	55	Pathogenic mutation
								11	c.3257G>A	c.3257G>A	p.Arg1086Gln)	Missense	1	33	VUS
								11	c.3447G>A	c.3447G>A	p.(Met1149Ile)	NA	1	39	VUS
								11	c.3578T>C	NA	NA	NA	1	33	NA
								11	c.4501G>A	NA	NA	NA	1	40	NA
								11	c.5206C>T (Q1736*)	c.5206C>T	p.(Gln1736Ter)	nonsense	1	30	Pathogenic mutation
								11	c.5104C>T (P1702S)	c.5104C>T	p.Pro1702Ser	missense	1	60	VUS
								18	c.8415G>T	c.8187G>T	p.Lys2729Asn	missense	1	40	Benign
								27	c.9976A>T (10204A>T)	c.9976A>T	p.Lys3326Ter	nonsense	1	52	Benign/ Likely Benign
								8	c.681+56C>T (IVS8+56C>T)	c.681+56C>T	Non coding	NA	1	54	Benign
								10	c.2137 + 22delT	NA	Intronic Variant	NA	1	33	Polymorphism
								24	c.9729-17T > C	NA	Non coding	NA	1	52	NA
Vidyanathan[27]	2009	39		India	BC	Familial BC	Direct sequencing	11	c.6078_6079del (6079delAGTT)	c.6078_6079del	p.(Glu2028ArgfsTer20)	Frameshift mutation	1	NA	Pathogenic mutation
Saxena S[23]	2002	20	25-64 years	India	BC	Familial BC	HDA/ PCR	18	c.8117A>G (8345A>G)	c.8117A>G	N2706S	missense, likely polymorphism	2	NA	VUS
								11	c.4779A>C (5007A>C)	c.4779A>C	E1593D	missense, likely polymorphism	1	NA	Benign
Soumittra[28]	2009	71	NA	India	BC	HBOC/HBC/Early BC	PCR-DHPLC	11	c.6214_6218delCTTAA	c.6214_6218delCTTAA	p.Ser2072SerfsX4	NA	1	38	NA
								11	c.5130_5133del (5357del4)	**c.5130_5133del**	p.Asp1709_Tyr1710insTer	Deletion	1	27	Pathogenic mutation
								11	c.2621_2627delAACTGTC	c.2621_2627del	p. Ile873IlefsX19	NA	1	35	NA

Note: Rashid et al. 2006 and Rashid et al. 2019 have BIC and HGVS nomenclature.

Abbreviations: BC- Breast cancer, HBOC- Hereditary Breast and Ovarian Cancer, HBC- Hereditary Breast Cancer, BIC- Breast Cancer Information Core, HGVS- Human Genome Variation Society, NGS - Next Gene Sequencing, PCR-Polymerase Chain reaction, SSCP-Single stranded Conformational Analysis, DHPLC - Denaturing High Performance Liquid Chromatography, HDA- Heteroduplex Analysis, DNA- Deoxy-Ribo Nucleic Acid, PTT- Protein Truncating Test